



peppwR: Simulation-Based Power Analysis for Phosphoproteomics Experiments in R

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Abstract

Phosphoproteomics experiments require careful experimental design to ensure adequate statistical power, yet no dedicated tools exist for power analysis that account for the unique statistical properties of peptide-level quantification data. We present **peppwR**, an R package for simulation-based power analysis tailored to phosphoproteomics. **peppwR** fits distributions to pilot data on a per-peptide basis, capturing the heterogeneity in abundance and variance across the peptidome. It supports multiple statistical tests (Wilcoxon, bootstrap-*t*, Bayes factor), tracks missingness patterns common in mass spectrometry data, and provides FDR-aware power calculations. Users can determine required sample sizes, estimate power for planned experiments, or identify minimum detectable effect sizes.

Keywords: power analysis, phosphoproteomics, mass spectrometry, experimental design, R.

1. Introduction

Statistical power—the probability of correctly rejecting a false null hypothesis—represents one of the most fundamental concepts in experimental design (Cohen 1988). Introduced by Neyman and Pearson (1933) and formalized by Cohen (1962), power analysis provides the theoretical framework for determining appropriate sample sizes, estimating the likelihood of detecting true effects, and identifying minimum detectable effect sizes. The consequences of inadequate power are profound: under-powered studies waste resources and risk missing genuine biological discoveries.

Despite some adoption in clinical trials (Schulz and Grimes 2005), power analysis remains systematically neglected in proteomics research. Most published proteomics studies do not report formal power calculations, in contrast to higher reporting rates observed in clinical research domains. This disparity persists despite the field’s evolution toward increasingly quantita-

tive approaches and the availability of sophisticated statistical methods for proteomics data analysis (Karp, Huber, Sadowski, Charles, Hester, and Lilley 2010).

Mass spectrometry-based proteomics presents unique statistical challenges that violate fundamental assumptions of traditional power analysis. Peptide abundances exhibit extreme heterogeneity, with concentrations spanning multiple orders of magnitude within a single experiment (Olsen, Blagoev, Gnäd, Macek, Kumar, Mortensen, and Mann 2006; Humphrey, Azimifar, and Mann 2015). This heterogeneity extends beyond mean abundances to encompass dramatic differences in variance structure across the peptidome (Karp *et al.* 2010). Abundance distributions can deviate substantially from normality, typically exhibiting right skewness that may be better characterized by alternative distributions such as gamma, log-normal, or inverse Gaussian models (Webb-Robertson, Wiber, Matzke, Samuel, Rodland, Clauss, and Pounds 2015). These distributional properties reflect both biological variation and the multiplicative nature of mass spectrometry quantification processes.

Missing values in proteomics follow systematic, non-random patterns that fundamentally alter statistical inference. Low-abundance peptides are disproportionately affected by detection limits, creating missing-not-at-random (MNAR) data structures (Webb-Robertson *et al.* 2015; Lazar, Gatto, Ferro, Bruley, and Burger 2016). The proportion of missing values can exceed 50% in discovery proteomics experiments (Tyanova, Temu, and Cox 2016a), with missingness rates negatively correlated with abundance levels—a pattern that violates the missing-completely-at-random (MCAR) assumptions underlying most statistical tests. Fourth, phosphoproteomics experiments routinely quantify thousands of peptides simultaneously, necessitating multiple testing corrections that substantially reduce effective statistical power (Karp *et al.* 2010; Zhang, Chambers, and Tabb 2018).

Current power analysis tools address these proteomics-specific challenges incompletely or not at all (Table 1). **Perseus** (Tyanova, Temu, Sinitcyn, Carlson, Hein, Geiger, Mann, and Cox 2016b), a widely-used analysis platform in proteomics, provides comprehensive statistical analysis including t-tests, ANOVA, and volcano plots, but offers no power analysis functionality. Users must rely on generic tools or ad-hoc approaches for experimental design decisions. **MSstats** (Chang, Picotti, Huttenhower, and Vitek 2014) provides sophisticated variance component modeling through linear mixed models and supports analysis at both peptide and protein levels, but utilizes aggregate variance components from fitted models rather than individual per-peptide distribution modeling needed for comprehensive phosphoproteomics experimental design. The **clippda** package (Nyangoma, Collins, Altman, Johnson, and Billingham 2012) was specifically designed for SELDI-TOF peak data and incorporates routines for handling data heterogeneities, but remains primarily focused on older mass spectrometry platforms rather than modern LC-MS/MS phosphoproteomics applications. **G*Power** (Faul, Erdfelder, Lang, and Buchner 2007), while widely used across scientific disciplines with over 30,000 citations, assumes normality and homogeneous variance, making it inappropriate for proteomics applications where data violate these assumptions. **MultiPower** (Tarazona, Balzano-Nogueira, Gómez-Cabrero, Schmidt, Madeira, and Conesa 2020) addresses power analysis in multi-omics contexts and is designed for integrative analysis across multiple platforms (genomics, transcriptomics, metabolomics), with limited focus on the specific statistical challenges of proteomics-only experiments. The tool focuses on pathway-level analysis rather than individual protein or peptide quantification.

The absence of tools that combine per-peptide distribution modeling with realistic missing data patterns and FDR-aware calculations has led researchers to either skip power analysis

entirely or apply inappropriate methods, potentially contributing to reproducibility issues and missed conclusions. This methodological gap is particularly problematic given the high costs associated with proteomics experiments and the limited sample availability in many contexts.

peppwR is an R package that directly addresses these limitations through simulation-based power analysis tailored specifically to mass spectrometry proteomics data. **peppwR** implements per-peptide distribution fitting to capture abundance heterogeneity, supports multiple statistical tests appropriate for non-normal data, incorporates realistic missing data patterns, and provides FDR-aware power calculations for high-throughput experiments. The package operates in two complementary modes: aggregate analysis for preliminary design decisions without pilot data, and per-peptide analysis that leverages pilot data to provide detailed, peptide-specific power estimates.

peppwR answers the three fundamental experimental design questions: “What sample size do I need to achieve target power?”, “What statistical power will my planned experiment achieve?”, and “What is the minimum detectable effect size given my sample size constraints?” By addressing the unique statistical properties of proteomics data, **peppwR** enables more informed experimental design decisions and contributes to improving the reproducibility and efficiency of proteomics research.

Table 1: Comparison of power analysis tools for proteomics applications.

Tool	Proteomics Focus	Per-feature Modeling	Non-normal Distributions	Missing Data Handling	Multiple Testing	Statistical Tests
peppwR	x	x	x	x	x	Wilcoxon, Bootstrap- <i>t</i> Bayes Factor
MSstats	x	¹	x	x	x	Linear mixed models
Perseus	x	—	x	x	x	t-test, ANOVA Volcano plots
clippda	²	—	—	—	—	t-test
MultiPower	³	x	—	—	x	Various
G*Power	—	—	—	—	—	t-test, ANOVA Correlation

2. Methods and Implementation

peppwR implements simulation-based power analysis through a modular framework that addresses the unique statistical challenges of phosphoproteomics data. The package operates in two complementary modes: aggregate analysis for preliminary design decisions without pilot data, and per-peptide analysis that leverages pilot data to provide detailed, peptide-specific power estimates across the heterogeneous peptidome.

2.1. Statistical Framework

Statistical power, defined as $P(\text{reject } H_0 | H_1 \text{ true}) = 1 - \beta$ (where β is the Type II error rate), provides the theoretical foundation for experimental design decisions. For a given statistical test with significance level α , power depends on the effect size δ , sample size n , and the underlying population distributions. In the context of proteomics experiments comparing

two groups, we test the null hypothesis $H_0 : \mu_{\text{treatment}} = \mu_{\text{control}}$ against the alternative $H_1 : \mu_{\text{treatment}} \neq \mu_{\text{control}}$.

peppwR accommodates three fundamental experimental design questions through constrained optimization. When `find = "sample_size"`, the package determines the minimum n required to achieve target power π_{target} :

$$n^* = \min\{n : P(\text{reject}H_0|\delta, n) \geq \pi_{\text{target}}\} \quad (1)$$

When `find = "power"`, it estimates the expected power for a fixed sample size:

$$\pi(n, \delta) = P(\text{reject}H_0|\delta, n) \quad (2)$$

When `find = "effect_size"`, it identifies the minimum detectable effect size:

$$\delta^* = \min\{\delta : P(\text{reject}H_0|\delta, n) \geq \pi_{\text{target}}\} \quad (3)$$

The effect size is parameterized as a multiplicative factor δ such that treatment group abundances are scaled by δ relative to control abundances. This multiplicative parameterization naturally accommodates the log-scale nature of mass spectrometry quantification, where biological effects typically manifest as fold-changes rather than additive differences.

In per-peptide mode, power calculations are performed independently for each peptide i with fitted distribution parameters θ_i , yielding peptide-specific power estimates $\pi_i(n, \delta)$. The package then reports the proportion of peptides achieving target power:

$$P_{\text{peptidome}}(n, \delta, \pi_{\text{target}}) = \frac{1}{K} \sum_{i=1}^K \mathbf{1}[\pi_i(n, \delta) \geq \pi_{\text{target}}] \quad (4)$$

where K is the total number of peptides and $\mathbf{1}[\cdot]$ is the indicator function.

For FDR-aware analysis, power is evaluated after Benjamini-Hochberg correction. Let $p_{(1)} \leq p_{(2)} \leq \dots \leq p_{(m)}$ denote the ordered p-values from m tests. The FDR-controlling procedure rejects all hypotheses $H_{(i)}$ for $i \leq k^*$, where:

$$k^* = \max \left\{ i : p_{(i)} \leq \frac{i \cdot \alpha}{m} \right\} \quad (5)$$

Power under FDR correction becomes the expected proportion of true positives among discoveries, accounting for the adaptive threshold imposed by the multiple testing procedure.

2.2. Distribution Fitting

peppwR models peptide abundance distributions using maximum likelihood estimation across multiple parametric families. The core distributions supported are gamma ($\text{Gamma}(\alpha, \beta)$), lognormal ($\text{LogNormal}(\mu, \sigma^2)$), normal ($\text{Normal}(\mu, \sigma^2)$), and inverse Gaussian ($\text{InverseGaussian}(\mu, \lambda)$). These distributions capture the right-skewed, positive-valued nature of mass spectrometry intensities while accommodating varying degrees of skewness and heteroscedasticity observed across the peptidome.

For each peptide i with observed abundances $\mathbf{x}_i = \{x_{i1}, x_{i2}, \dots, x_{in_i}\}$, the package fits each candidate distribution with probability density function $f_k(x|\theta_k)$ by maximizing the log-likelihood:

$$\hat{\theta}_{k,i} = \arg \max_{\theta_k} \sum_{j=1}^{n_i} \log f_k(x_{ij}|\theta_k) \quad (6)$$

Model selection employs the Akaike Information Criterion (AIC), selecting the distribution with minimum AIC score:

$$\hat{k}_i = \arg \min_k \left\{ -2 \sum_{j=1}^{n_i} \log f_k(x_{ij}|\hat{\theta}_{k,i}) + 2p_k \right\} \quad (7)$$

where p_k is the number of parameters in distribution k .

The gamma distribution, parameterized by shape $\alpha > 0$ and rate $\beta > 0$, provides flexibility in modeling right-skewed abundance data with controlled variance-to-mean relationships. The lognormal distribution naturally accommodates multiplicative biological processes underlying protein expression, while the inverse Gaussian distribution offers an intermediate option between gamma and lognormal with distinct tail behavior. Additional candidate distributions including Weibull, exponential, and beta (for transformed data) are available through the **univariateML** interface.

Parameter estimation leverages the **fitdistrplus** package for standard distributions and **univariateML** for specialized cases. Both packages implement numerically stable algorithms with appropriate handling of boundary conditions and convergence diagnostics. When maximum likelihood estimation fails due to insufficient data, extreme parameter estimates, or numerical instability, **peppwR** provides three fallback strategies controlled by the `on_fit_failure` argument.

The **"exclude"** strategy removes problematic peptides from analysis, reporting the exclusion count for transparency. The **"empirical"** strategy substitutes parametric distributions with empirical bootstrap resampling from observed values, preserving the exact empirical distribution at the cost of reduced statistical efficiency. The **"lognormal"** strategy fits a moment-matched lognormal distribution using method-of-moments estimates, providing a robust fallback when maximum likelihood fails but ensuring all peptides remain in analysis.

Distribution fitting diagnostics include goodness-of-fit statistics (Kolmogorov-Smirnov, Anderson-Darling), QQ plots comparing theoretical and empirical quantiles, and density overlay plots visualizing the fitted distribution against observed data histograms. These diagnostics enable users to assess the adequacy of distributional assumptions and identify peptides where parametric modeling may be inappropriate.

2.3. Simulation Engine

Power estimation employs Monte Carlo simulation to accommodate the complex statistical properties of proteomics data that preclude analytical power calculations. The simulation framework generates synthetic datasets under specified experimental conditions, applies statistical tests, and estimates power as the proportion of simulations achieving statistical significance.

For each peptide i with fitted distribution $f_i(\cdot|\hat{\theta}_i)$ and each candidate sample size n , the simulation algorithm proceeds as follows:

1. Generate n control group samples: $X_{i,\text{control}}^{(s)} \sim f_i(\cdot|\hat{\theta}_i)$ for $s = 1, \dots, n$
2. Generate n treatment group samples: $X_{i,\text{treatment}}^{(s)} \sim f_i(\cdot|\delta \cdot \hat{\theta}_i)$ for $s = 1, \dots, n$, where δ represents the multiplicative effect size
3. Apply missing data patterns if `include_missingness = TRUE`, randomly removing observations according to the peptide-specific missingness rate r_i
4. Compute the test statistic and p-value using the specified statistical test
5. Record rejection outcome: $R^{(j)} = \mathbf{1}[p^{(j)} \leq \alpha]$ for simulation j
6. Repeat for n_{sim} independent simulations

Power is estimated as:

$$\hat{\pi}_i(n, \delta) = \frac{1}{n_{\text{sim}}} \sum_{j=1}^{n_{\text{sim}}} R^{(j)} \quad (8)$$

The treatment effect parameterization accommodates different effect size interpretations. For location-scale families, the effect size δ scales the location parameter (mean for normal, rate parameter for gamma, location parameter for lognormal). This multiplicative scaling preserves the distributional shape while shifting the central tendency, consistent with biological interpretation of fold-changes in proteomics.

The default $n_{\text{sim}} = 1000$ simulations provides adequate precision for most applications, but can be increased for higher precision requirements.

The simulation engine supports vectorized operations across peptides and sample sizes, enabling efficient computation of power curves and sample size determinations. Parallel processing capabilities allow distribution across multiple CPU cores when analyzing large peptidomes, with linear scaling up to the number of available cores.

For FDR-aware simulations, the engine generates complete experiments across all peptides simultaneously, applies the Benjamini-Hochberg procedure to the resulting p-value vectors, and computes power as the proportion of true positives among discoveries. This approach correctly accounts for the dependence structure introduced by multiple testing correction, providing realistic power estimates for high-throughput experiments.

2.4. Missing Data Handling

peppwR incorporates missing data mechanisms specific to mass spectrometry-based proteomics, where detection limits create systematic patterns of missing-not-at-random (MNAR) data. The package implements a two-level approach to characterizing and incorporating missingness patterns into power calculations.

At the dataset level, **peppwR** detects MNAR patterns by computing the Spearman correlation between peptide-level mean abundances and missingness rates. A significant negative correlation ($\rho < -0.3$, $p < 0.05$) indicates that low-abundance peptides are systematically more likely to be missing—the signature of detection-limit-driven MNAR common in discovery proteomics (Webb-Robertson *et al.* 2015). This dataset-level diagnostic provides users with quantitative evidence for MNAR mechanisms and informs the interpretation of power analysis results.

The MNAR detection algorithm computes peptide-specific summary statistics:

$$\bar{x}_i = \frac{1}{n_{\text{obs},i}} \sum_{j: x_{ij} \neq \text{NA}} x_{ij} \quad (9)$$

$$r_i = \frac{n_{\text{missing},i}}{n_{\text{total},i}} \quad (10)$$

where $\bar{x}_i$ is the mean abundance for peptide i (excluding missing values), r_i is the missingness rate, $n_{\text{obs},i}$ is the number of observed values, and $n_{\text{missing},i}$ is the number of missing values.

At the peptide level, the package records the observed missingness rate r_i for each peptide during the distribution fitting phase. When `include_missingness = TRUE` is specified in power analysis, simulations incorporate these peptide-specific NA rates by randomly removing observations from generated samples with probability r_i . This approach preserves the heterogeneous pattern of data loss across peptides while maintaining the stochastic nature of missing data generation.

The missing data incorporation algorithm modifies the standard simulation procedure:

1. Generate complete control and treatment samples as described above
2. For each peptide i , randomly select $\lfloor n \cdot r_i \rfloor$ observations for removal from each group
3. Apply the specified statistical test to the resulting incomplete data
4. Record the test outcome, including cases where insufficient data remains for testing

When missingness results in insufficient data for statistical testing (typically when fewer than 3 observations remain in either group), the peptide is excluded from that simulation iteration. The proportion of successful tests is reported alongside power estimates, providing transparency about the impact of missing data on statistical inference.

This missingness-aware simulation provides realistic power estimates that account for expected data loss, enabling more accurate sample size determinations for experiments anticipated to have substantial missing values. Users can compare power estimates with and without missingness incorporation to quantify the impact of incomplete data on experimental design requirements.

2.5. Software Architecture

peppwR implements an S3 object-oriented architecture with two primary classes: `'peppwr_fits'` and `'peppwr_power'`, each equipped with specialized methods for printing, plotting, and summarization.

The `'peppwr_fits'` class encapsulates distribution fitting results, storing fitted parameters, goodness-of-fit statistics, and metadata for each peptide. The class structure includes:

- **data**: Original input data as nested tibble
- **fits**: List of fit result tibbles per peptide containing distribution names, log-likelihood values, and AIC scores
- **best**: Character vector indicating the best-fitting distribution for each peptide

- `call`: Original function call for reproducibility
- `missingness`: Tibble with peptide-level missingness rates and diagnostic statistics (optional)
- `dataset_mnar`: Dataset-level MNAR correlation metric and interpretation (optional)

The `'peppwr_power'` class stores power analysis results with associated simulation parameters:

- `mode`: Analysis mode ("aggregate" or "per_peptide")
- `question`: The solved question ("power", "sample_size", or "effect_size")
- `answer`: The computed result value
- `simulations`: List containing simulation results and configuration details
- `params`: List of input parameters used in the analysis
- `call`: Original function call for reproducibility

Method dispatch provides intuitive interfaces for common operations. The `print()` methods deliver concise summaries appropriate for interactive analysis, while `summary()` methods provide detailed statistical output including confidence intervals and fit diagnostics. The `plot()` methods generate diagnostic visualizations tailored to each class.

The design separates core functionality into distinct components: distribution fitting (`fit_distributions()`), power analysis (`power_analysis()`), simulation engine (`run_power_sim()`), statistical tests (`test_wilcoxon()`, `test_bootstrap_t()`, `test_bayes_t()`), and visualization (`plot.peppwr_fits()`, `plot.peppwr_power()`). This modular architecture enables flexible extension while maintaining clear separation of concerns.

3. Examples

3.1. Aggregate Mode Analysis

The **peppwR** package implements two complementary analysis modes. Aggregate mode operates with user-specified distribution parameters rather than pilot data, providing rapid power estimates for experimental planning scenarios where pilot data is unavailable. This mode addresses the three fundamental power analysis questions through Monte Carlo simulation using parametric distributions.

Distribution Parameterization

Aggregate mode requires specification of a distribution family and associated parameters. The package accepts parameters as named lists corresponding to standard R distribution parameterizations.

```
R> library(peppwR)
```


For demonstration purposes, we use gamma distribution parameters with shape values ranging from 1.5 to 5 and rate parameters from 0.01 to 0.1. Lower shape parameters generate more skewed distributions with higher variance, while higher shape parameters produce more symmetric distributions with reduced variance.

Sample Size Determination

The `power_analysis()` function determines required sample sizes through iterative simulation. When `find = "sample_size"`, the function identifies the minimum per-group sample size necessary to achieve a specified target power for a given effect size.

```
R> set.seed(123)
R> result <- power_analysis(
+   distribution = "gamma",
+   params = list(shape = 2, rate = 0.05),
+   effect_size = 2,
+   target_power = 0.8,
+   find = "sample_size",
+   n_sim = 1000
+ )
R> print(result)
```

```
peppwr_power analysis
```

```
-----
```

```
Mode: aggregate
```

```
Recommended sample size: N=25 per group
```

```
Target power: 80%
```

```
Effect size: 2.00-fold
```

```
Statistical test: wilcoxon
```

```
Significance level: 0.05
```

The algorithm performs 1000 Monte Carlo simulations to identify the minimum sample size achieving the target power of 80%, determining that 25 samples per group are required (Figure 1).

```
R> plot(result)
```

Figure 1 demonstrates how statistical power increases with sample size, with the recommended value marked at the intersection with the target power threshold.

Power Estimation

Power estimation for fixed experimental designs operates through the `find = "power"` specification. This mode simulates experiments with predetermined sample sizes and effect sizes to estimate the probability of correctly rejecting false null hypotheses.

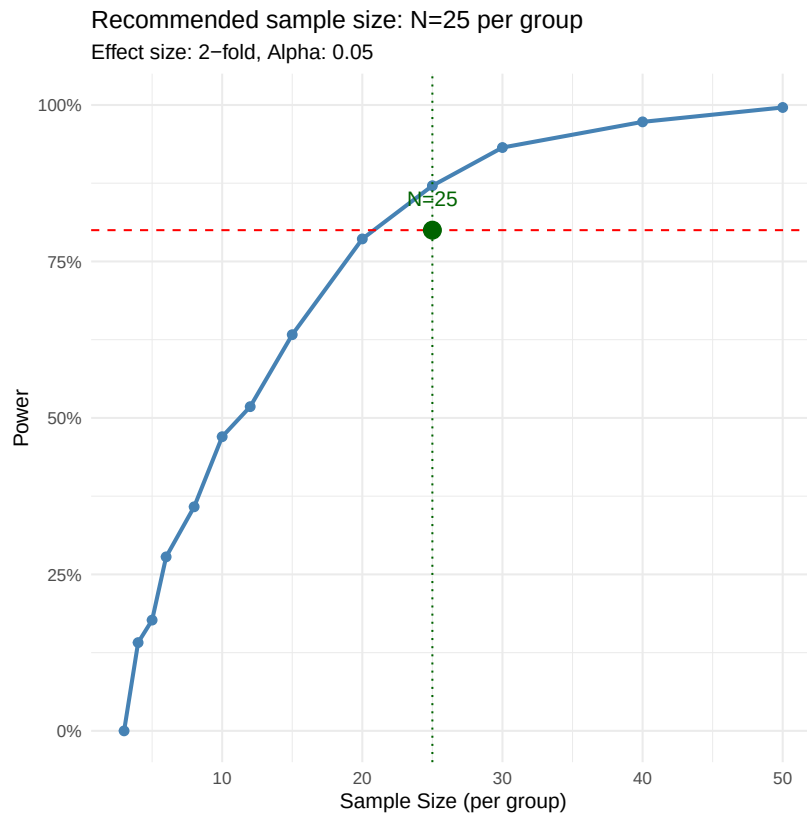


Figure 1: Power curve showing the relationship between sample size and achieved power. The recommended sample size corresponds to the first value achieving the target power of 0.8.

```
R> set.seed(123)
R> result <- power_analysis(
+   distribution = "gamma",
+   params = list(shape = 2, rate = 0.05),
+   effect_size = 2,
+   n_per_group = 6,
+   find = "power",
+   n_sim = 1000
+ )
R> print(result)
```

```
peppwr_power analysis
```

```
-----
Mode: aggregate
```

```
Power: 24%
```

```
Sample size: 6 per group
```

```
Effect size: 2.00-fold
```

```
Statistical test: wilcoxon
```

Significance level: 0.05

The simulation reveals that 6 samples per group provide 24% power to detect a 2-fold effect under the specified distributional assumptions, substantially below the conventional 80% power threshold.

Minimum Detectable Effect Size

The package determines minimum detectable effect sizes through `find = "effect_size"` specifications. This analysis identifies the smallest multiplicative effect that can be reliably detected given sample size and power constraints (Figure 2).

```
R> set.seed(123)
R> result <- power_analysis(
+   distribution = "gamma",
+   params = list(shape = 2, rate = 0.05),
+   n_per_group = 6,
+   target_power = 0.8,
+   find = "effect_size",
+   n_sim = 1000
+ )
R> print(result)
```

```
peppwr_power analysis
```

```
-----
```

```
Mode: aggregate
```

```
Minimum detectable effect: 5.00-fold
```

```
Sample size: 6 per group
```

```
Target power: 80%
```

```
Statistical test: wilcoxon
```

```
Significance level: 0.05
```

```
R> plot(result)
```

As shown in Figure 2, the analysis indicates that 6 samples per group can reliably detect fold-changes of 5.0 or greater with 80% power, but smaller effects would require larger sample sizes for adequate detection probability.

Parameter Sensitivity Analysis

Aggregate mode facilitates sensitivity analysis by varying distributional assumptions. Different shape parameters reflect varying degrees of heterogeneity and skewness in abundance distributions.

```
R> set.seed(123)
R>
```

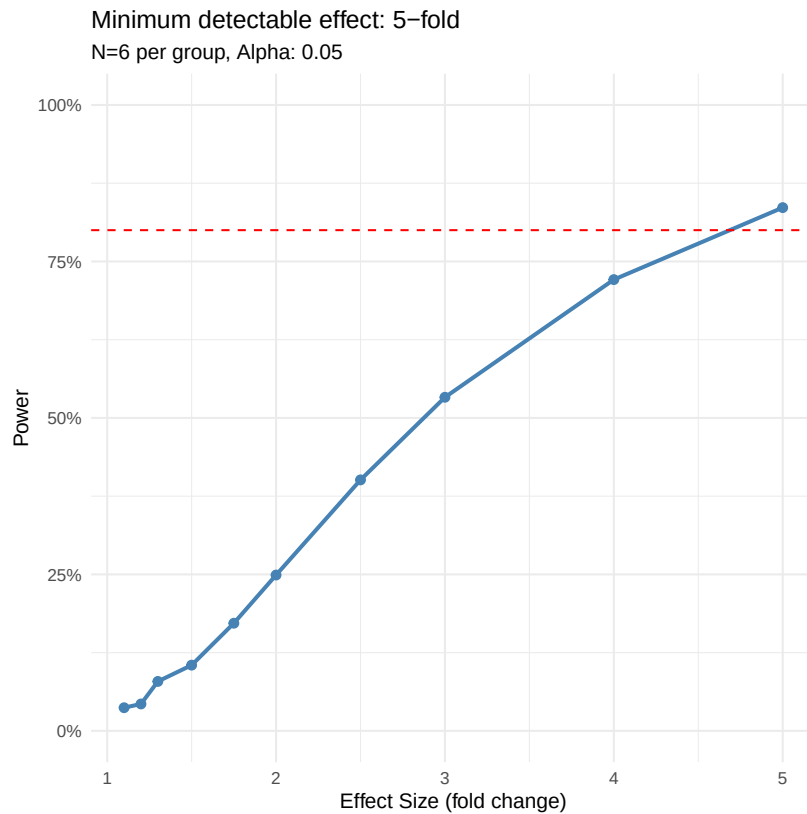


Figure 2: Power as a function of effect size for fixed sample size (N=6 per group). The minimum detectable effect size represents the smallest fold-change reliably detectable with 80% power.

```
R> # Conservative (high variance) scenario
R> conservative <- power_analysis(
+   "gamma",
+   params = list(shape = 1.5, rate = 0.05),
+   effect_size = 2,
+   target_power = 0.8,
+   find = "sample_size",
+   n_sim = 500
+ )
R>
R> # Moderate variance scenario
R> moderate <- power_analysis(
+   "gamma",
+   params = list(shape = 2.5, rate = 0.05),
+   effect_size = 2,
+   target_power = 0.8,
+   find = "sample_size",
+   n_sim = 500
+ )
```

```

+ )
R>
R> # Optimistic (low variance) scenario
R> optimistic <- power_analysis(
+   "gamma",
+   params = list(shape = 4, rate = 0.05),
+   effect_size = 2,
+   target_power = 0.8,
+   find = "sample_size",
+   n_sim = 500
+ )
R>
R> cat("Conservative (shape=1.5): N =", conservative$answer, "per group\n")
R> cat("Moderate (shape=2.5):      N =", moderate$answer, "per group\n")
R> cat("Optimistic (shape=4):      N =", optimistic$answer, "per group\n")

Conservative (shape=1.5): N = 30 per group
Moderate (shape=2.5):      N = 20 per group
Optimistic (shape=4):      N = 12 per group

```

The sensitivity analysis demonstrates how distributional assumptions substantially affect sample size requirements. More skewed distributions (lower shape parameters) require larger sample sizes to achieve equivalent power, reflecting the increased variance associated with heavily skewed abundance patterns. This analysis provides bounds for experimental planning under uncertainty about the true underlying distributions.

Acknowledgments

All acknowledgments (note the AE spelling) should be collected in this unnumbered section before the references. It may contain the usual information about funding and feedback from colleagues/reviewers/etc. Furthermore, information such as relative contributions of the authors may be added here (if any).

References

- Chang CY, Picotti P, Huttenhower C, Vitek O (2014). “**MSstats**: An R Package for Statistical Analysis of Quantitative Mass Spectrometry-based Proteomic Experiments.” *Bioinformatics*, **30**(17), 2524–2526. doi:10.1093/bioinformatics/btu305.
- Cohen J (1962). “The statistical power of abnormal-social psychological research: a review.” *The Journal of Abnormal and Social Psychology*, **65**(3), 145–153. doi:10.1037/h0045186.
- Cohen J (1988). *Statistical Power Analysis for the Behavioral Sciences*. 2nd edition. Lawrence Erlbaum Associates. doi:10.4324/9780203771587.

- Faul F, Erdfelder E, Lang AG, Buchner A (2007). “**G*Power** 3: A Flexible Statistical Power Analysis Program for the Social, Behavioral, and Biomedical Sciences.” *Behavior Research Methods*, **39**(2), 175–191. doi:10.3758/bf03193146.
- Humphrey SJ, Azimifar SB, Mann M (2015). “High-throughput phosphoproteomics reveals in vivo insulin signaling dynamics.” *Nature Biotechnology*, **33**(9), 990–995. doi:10.1038/nbt.3327.
- Karp NA, Huber W, Sadowski PG, Charles PD, Hester SV, Lilley KS (2010). “Addressing accuracy and precision issues in iTRAQ quantitation.” *Molecular & Cellular Proteomics*, **9**(9), 1885–1897. doi:10.1074/mcp.M900628-MCP200.
- Lazar C, Gatto L, Ferro M, Bruley C, Burger T (2016). “Accounting for the multiple natures of missing values in label-free quantitative proteomics data sets to compare imputation strategies.” *Journal of Proteome Research*, **15**(4), 1116–1125. doi:10.1021/acs.jproteome.5b00981.
- Neyman J, Pearson ES (1933). “On the problem of the most efficient tests of statistical hypotheses.” *Philosophical Transactions of the Royal Society of London. Series A, Containing Papers of a Mathematical or Physical Character*, **231**(694-706), 289–337. doi:10.1098/rsta.1933.0009.
- Nyangoma SO, Collins SI, Altman DG, Johnson P, Billingham L (2012). “Sample Size Calculations for Designing Clinical Proteomic Profiling Studies Using Mass Spectrometry.” *Statistical Applications in Genetics and Molecular Biology*, **11**(3). doi:10.1515/1544-6115.1686. **clippda** R package available from Bioconductor.
- Olsen JV, Blagoev B, Gnäd F, Macek B, Kumar C, Mortensen P, Mann M (2006). “Global, in vivo, and site-specific phosphorylation dynamics in signaling networks.” *Cell*, **127**(3), 635–648. doi:10.1016/j.cell.2006.09.026.
- Schulz KF, Grimes DA (2005). “Sample size calculations in randomised trials: mandatory and mystical.” *The Lancet*, **365**(9467), 1348–1353. doi:10.1016/S0140-6736(05)61034-3.
- Tarazona S, Balzano-Nogueira L, Gómez-Cabrero D, Schmidt A, Madeira SC, Conesa A (2020). “Harmonization of Quality Metrics and Power Calculation in Multi-Omic Studies.” *Nature Communications*, **11**(1), 3092. doi:10.1038/s41467-020-16937-8.
- Tyanova S, Temu T, Cox J (2016a). “The MaxQuant computational platform for mass spectrometry-based shotgun proteomics.” *Nature Protocols*, **11**(12), 2301–2319. doi:10.1038/nprot.2016.136.
- Tyanova S, Temu T, Sinitcyn P, Carlson A, Hein MY, Geiger T, Mann M, Cox J (2016b). “The **Perseus** Computational Platform for Comprehensive Analysis of (Prote)omics Data.” *Nature Methods*, **13**(9), 731–740. doi:10.1038/nmeth.3901.
- Webb-Robertson BJM, Wiber HK, Matzke MM, Samuel JN, Rodland KD, Clauss TR, Pounds JG (2015). “Review, evaluation, and discussion of the challenges of missing value imputation for mass spectrometry-based label-free global proteomics.” *Journal of Proteome Research*, **14**(5), 1993–2001. doi:10.1021/pr501138h.

Zhang B, Chambers MC, Tabb DL (2018). “Multiple testing in proteomics: a comprehensive review.” *Molecular & Cellular Proteomics*, **6**(12), 2032–2041. doi:[10.1074/mcp.R700002-MCP200](https://doi.org/10.1074/mcp.R700002-MCP200).

A. More technical details

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- Many other potentially helpful details...

B. Using BibT_EX

References need to be provided in a BibT_EX file (`.bib`). All references should be made with `\cite`, `\citet`, `\citep`, `\citealp` etc. (and never hard-coded). These commands yield different formats of author-year citations and allow to include additional details (e.g., pages, chapters, ...) in brackets. In case you are not familiar with these commands see the JSS style FAQ for details.

Cleaning up BibT_EX files is a somewhat tedious task – especially when acquiring the entries automatically from mixed online sources. However, it is important that informations are complete and presented in a consistent style to avoid confusions. JSS requires the following format.

- JSS-specific markup (`\proglang`, `\pkg`, `\code`) should be used in the references.
- Titles should be in title case.
- Journal titles should not be abbreviated and in title case.
- DOIs should be included where available.
- Software should be properly cited as well. For R packages `citation("pkgname")` typically provides a good starting point.

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